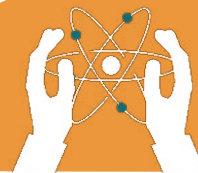


UNIT 6

ACIDS AND ALKALIS

Learning objectives:

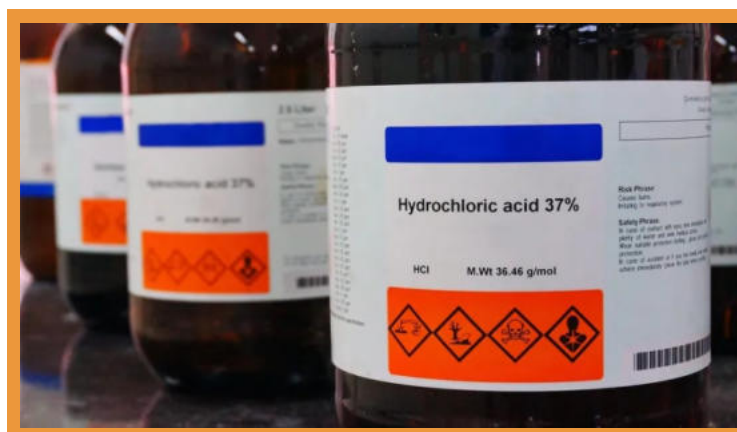
- Identify acids and alkalis, their properties, and common examples.
- Explain the pH scale and its use in determining acidity or alkalinity.
- Understand neutralization and its applications in daily life.



6.1 What Are Acids?

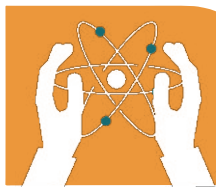
Acids are substances that taste sour and turn blue litmus paper red. They are commonly found in foods like lemons (**citric acid**) and vinegar (**acetic acid**). Acids are also used in industries for making fertilizers, medicines, and cleaning products. Acids can be strong or weak:

- **Strong acids:** Hydrochloric acid (HCl) and sulfuric acid (H_2SO_4).
- **Weak acids:** Citric acid and acetic acid.



6.2 What Are Alkalis?

Alkalis are substances that feel soapy and turn red litmus paper blue. They are a type of base that dissolves in water. Common alkalis include baking soda (sodium



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bicarbonate) and soaps. Alkalis are used in cleaning products, making soaps, and in cooking. Like acids, alkalis can also be strong or weak:

- **Strong alkalis:** Sodium hydroxide (NaOH).
- **Weak alkalis:** Baking soda and limewater.



Sodium Hydroxide
Soaps



Sodium Carbonate
Detergents



Bases used in
Labs



Calcium Hydroxide
Cement



Sodium Fluoride
Toothpaste



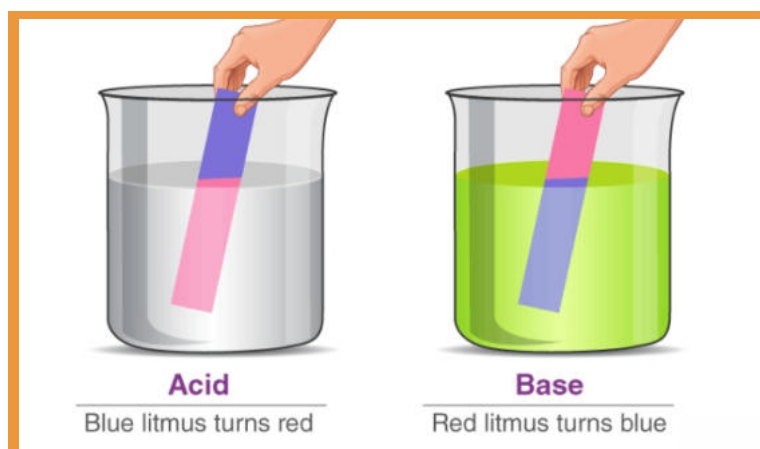
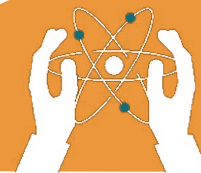
Ammonia
Fertilizers

6.3 What Are Indicators?

Indicators are special substances used to identify whether a substance is acidic or alkaline. They change colour based on the substance:

- **Litmus paper:** Blue litmus turns red in acid, and red litmus turns blue in alkali.
- **Natural indicators:** Turmeric (yellow turns red in alkali) and red cabbage juice.

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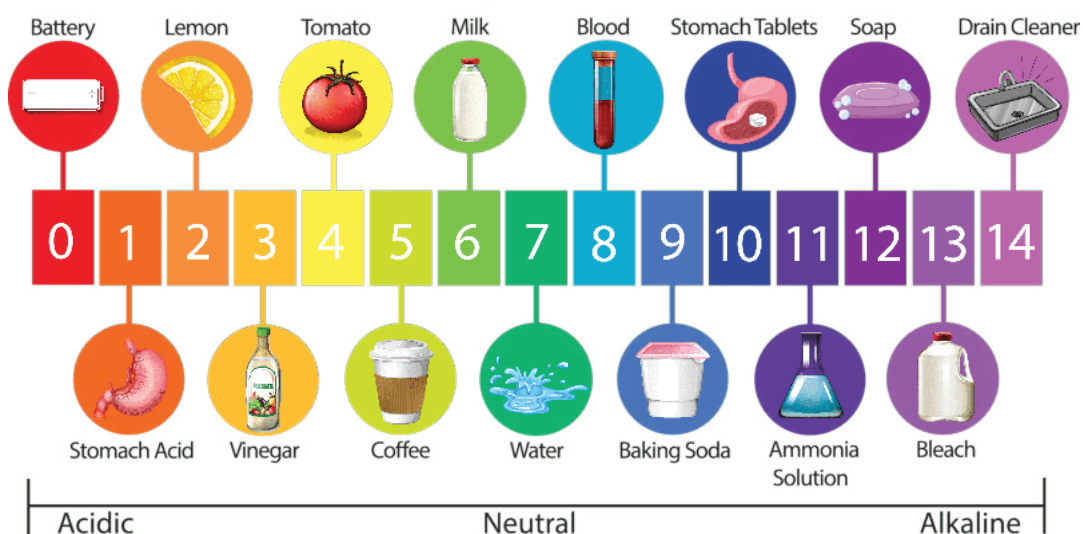


6.4 What Is the pH Scale?

The pH scale is used to measure how acidic or alkaline a substance is. The scale ranges from 0 to 14:

- Substances with a pH less than 7 are acidic.
- Substances with a pH greater than 7 are alkaline.
- A pH of 7 is neutral (e.g., water).

The pH Scale



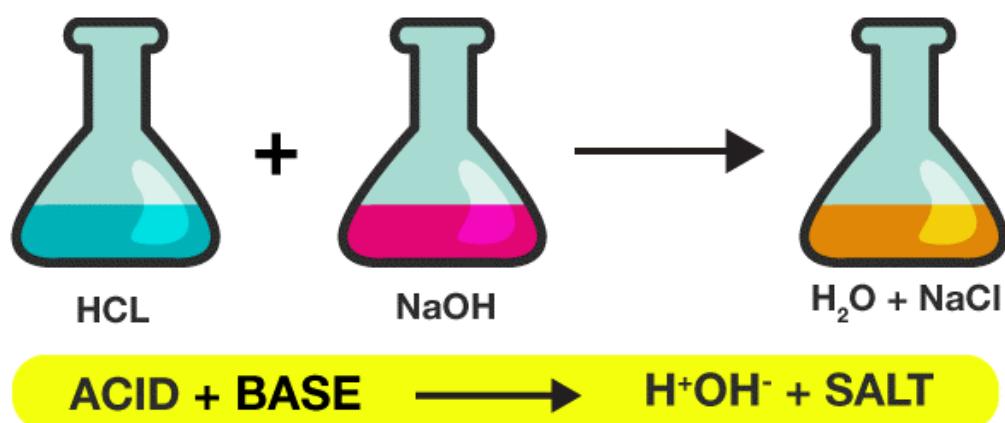


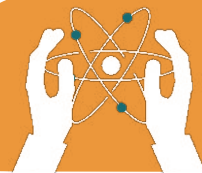
6.5 Neutralization

Neutralization is a chemical reaction where an acid and an alkali react to form a neutral substance, usually water and a salt. This reaction is used in everyday life, such as:

- It is used in treating indigestion (neutralizing stomach acid with antacids).
- It is used in treating acidic soil with lime.

They are used in neutralizing acids in chemical spills.





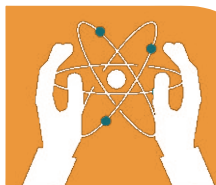
EXERCISES

Fill in the blanks.

1. Acids turn _____ litmus paper red.
2. Baking soda is an example of a _____.
3. A pH of 7 is considered _____.
4. Strong acids include _____ and sulfuric acid.
5. The pH scale ranges from _____ to _____.
6. Litmus paper is an example of an _____.
7. Neutralization forms a _____ and water.
8. Vinegar contains the acid called _____.

Tick(✓) for True and cross (x) for False statements

1. Alkalis turn red litmus paper blue.
2. All acids are strong.
3. Lemon juice contains acetic acid.
4. Sodium hydroxide is a weak alkali.
5. A substance with a pH of 6 is acidic.
6. Neutralization produces harmful substances.
7. Litmus paper is used to test for acids and alkalis.
8. The pH of water is 7.



Choose the correct answer

1. The pH of a strong alkali is:

- a) Less than 7 b) Equal to 7 c) Greater than 7 d) Near 14

3. An example of an acid is:

- a) Soap b) Toothpaste c) Baking soda d) Vinegar

3. Neutralization produces:

- a) Salt and water b) Acid and alkali c) Gas d) Precipitate

4. The pH of a neutral substance is:

- a) 5 b) 7 c) 10 d) 14

5. Which is a natural indicator?

- a) Baking soda b) Methyl orange c) Sodium hydroxide
d) Red cabbage juice

6. Sodium hydroxide is a:

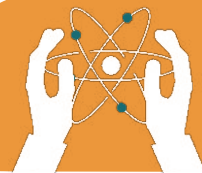
- a) Strong acid b) Strong alkali c) Weak acid
d) Neutral substance

7. What turns red in an alkali?

- a) Blue litmus b) Red litmus c) Both d) None

8. What do acids usually taste like?

- a) Sweet b) Sour c) Salty d) Bitter



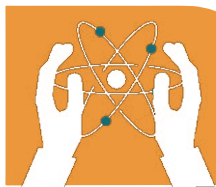
Answer the Following Questions:

Short-Answer Questions

1. What are Alkalis? Write two examples and uses.
2. What does the pH scale measure?
3. What are indicators? How do they change their colour in different substances?
4. Define neutralization and its uses.

Long-Answer Questions

1. Explain the pH scale with examples of substances at different pH values.
2. What are acids? Explain the difference between strong and week acid?



WORKSHEET

Acids and Alkalies

1. A pH probe can be used to get an accurate reading from different liquids. An example of a pH probe is shown below.



A variety of solutions were tested, and the results are shown in the table below. Put a ✓ in the correct box to show whether the solution was acidic, alkaline, or neutral. (5+5+2)

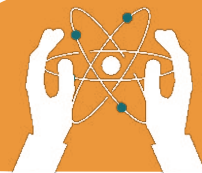
Solution	pH Reading	Acidic	Alkaline	Neutral
Lemon juice	2.2			
Toothpaste	9.9			
Blood	7.4			
Pure water	7.0			
Tomato juice	4.4			

Questions:

1. Give a reason why using a pH probe is a better way of determining a solution's pH rather than using a universal indicator.

2. Between each test, the probe should be dipped in pure water. Suggest a reason why.

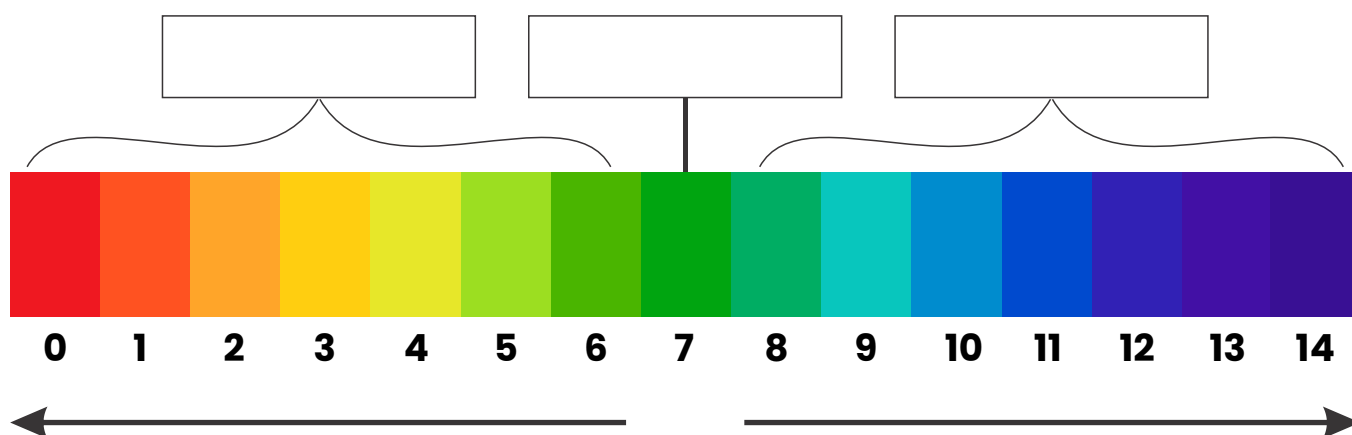
ACIDS AND ALKALIS



3. What is an indicator?

4. Name 2 other indicators that could have been used.

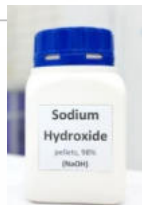
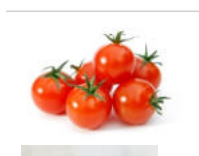
1. The PH scale is used to measure how acidic or alkaline a substance is. Fill in the blanks below:



The _____ the number,
the _____ the acid

The _____ the number,
the _____ the alkali

2. Match the pictures with the columns.



Acid	Alkali

3. What is a characteristic of an acid and an alkali:

Acid _____

Alkali _____